

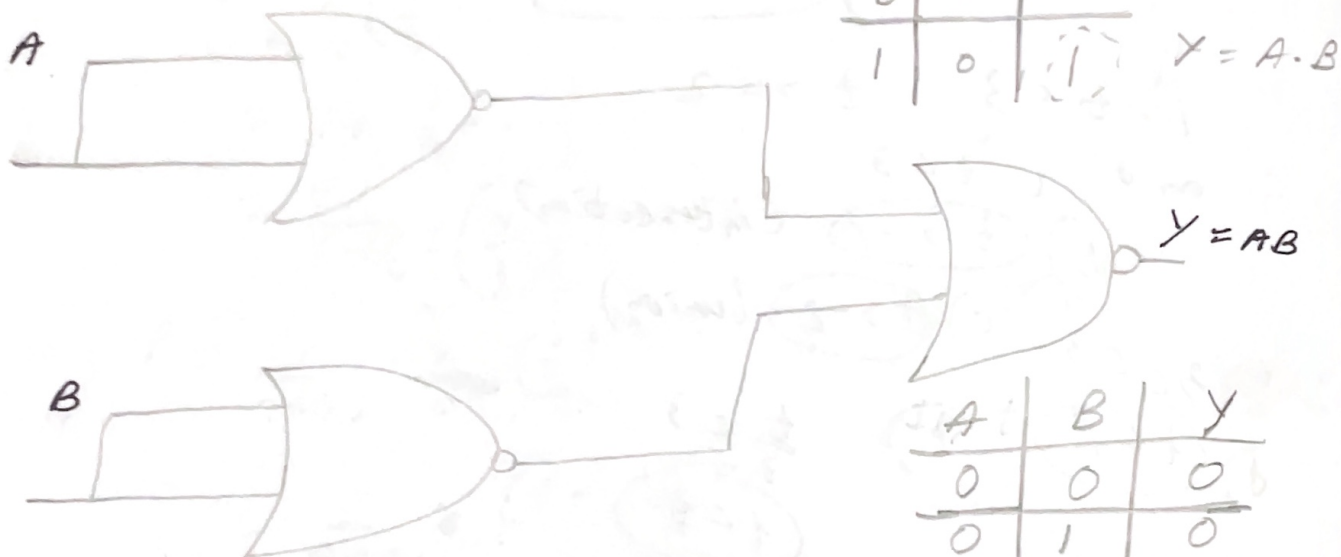
EE214 Lab-1 Report

Name: Medhansh
Roll No.: 25B3986

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Roll No.: 25B3986

Date: 7/8/26

i) AND gate using NOR gate(s)

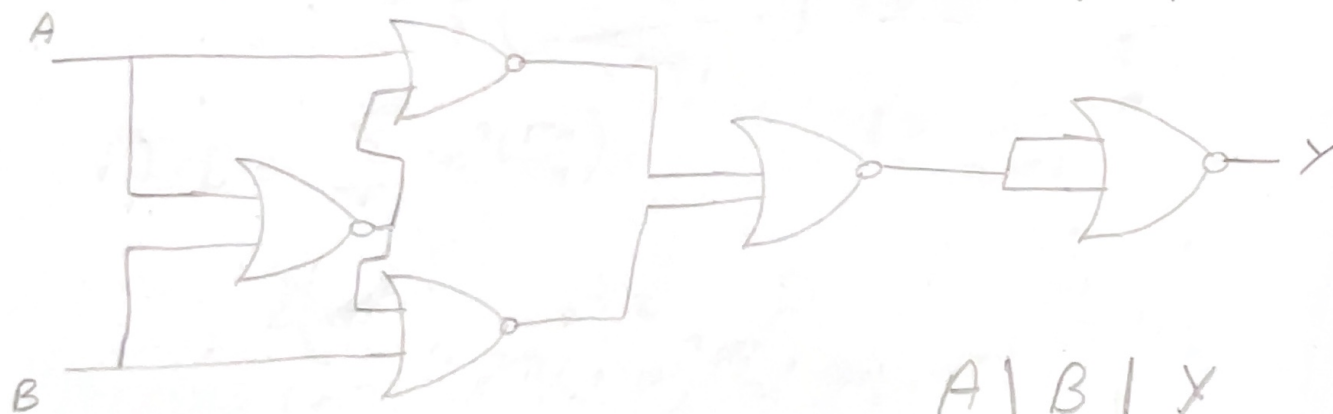


A \ B	0	1
0	0	0
1	0	1

$$Y = A \cdot B$$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

ii) XOR gate using NOR gate(s)

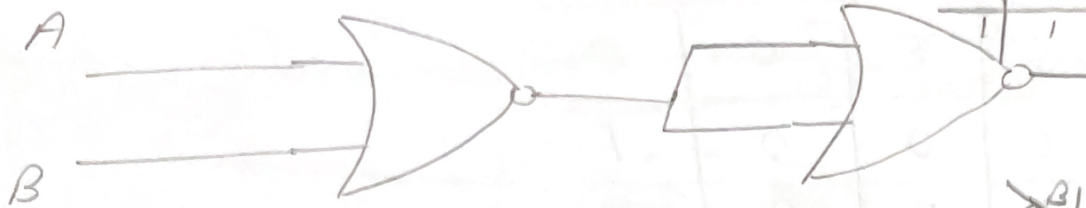


A \ B	0	1
0	0	1
1	1	0

$$Y = A \oplus B$$

A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0

iii) OR gate using NOR gate(s)



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

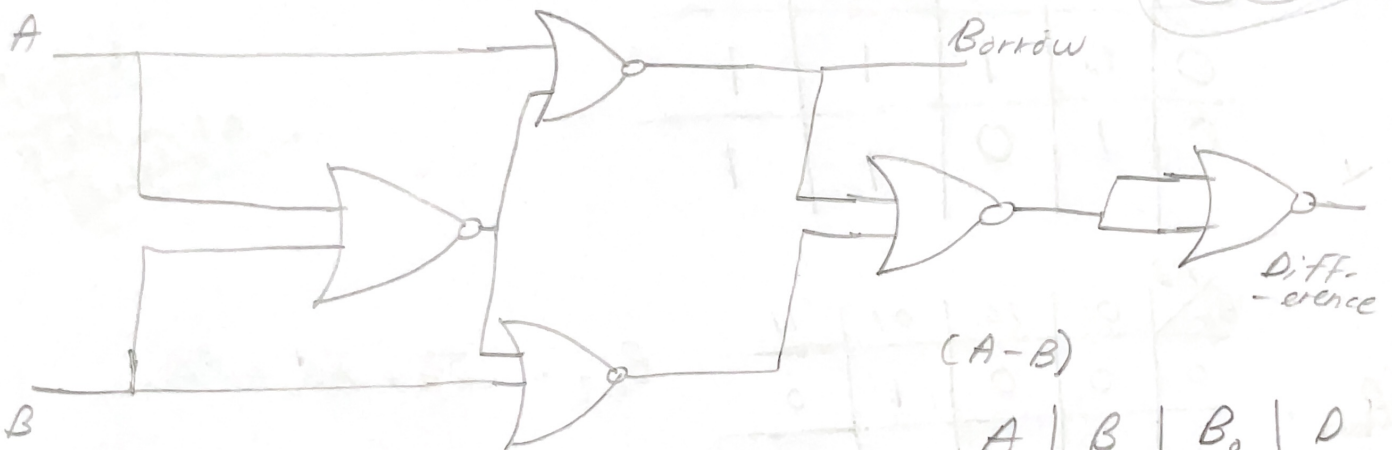
0	0	1
0	0	1
1	1	1

$$Y = A + B$$

iv) Half subtractor using NOR gate(s)

Difference $\Rightarrow$ XOR ($A \oplus B$)

Borrow $\Rightarrow \bar{A} \cdot B$



$A \cap B = \overline{A \cup B}$



A	B	B_{in}	B_{out}	D
0	0	0	0	0
1	0	0	0	1
1	1	0	0	0
1	0	1	0	0
1	1	1	1	1
0	1	1	1	0
0	0	1	1	1
0	1	0	1	1

(iii) OR gate using NOR gate (2)

(iv) Half subtractor using NOR gate (2)
 Difference = $A \oplus B$
 Borrow = $\overline{A \cdot B}$

$B_{out} :$

$B_{in} \backslash AB$	00	10	01	11
0	0	0	1	0
1	1	0	1	1

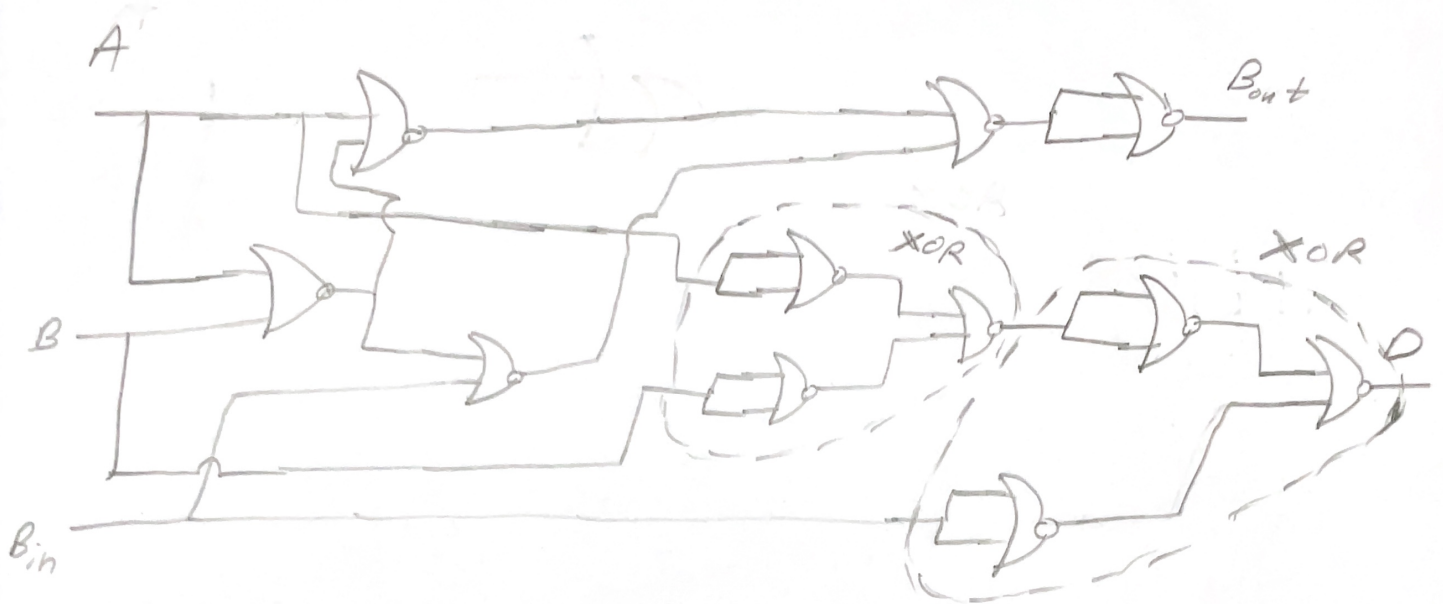
$$B_{out} = \overline{A}B + \overline{A}B_{in} + AB_{in}$$

D:

$B_{in} \backslash AB$	00	10	01	11
0	0	1	1	0
1	1	0	0	1

(2) Full subtractor using NOR gate (4)
 $D = A \oplus B \oplus B_{in}$

Rather,



Verified
7/8/20
Detholp

OR gate

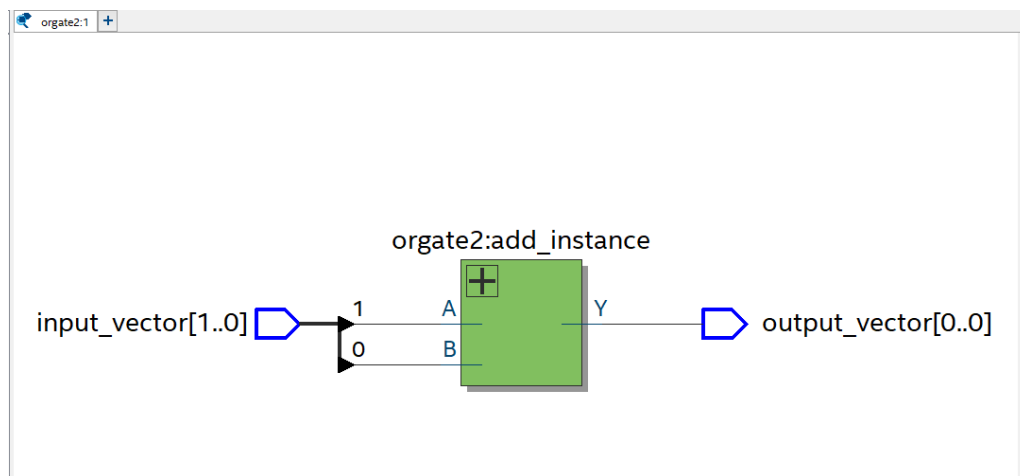


Figure 1: OR gate schematic

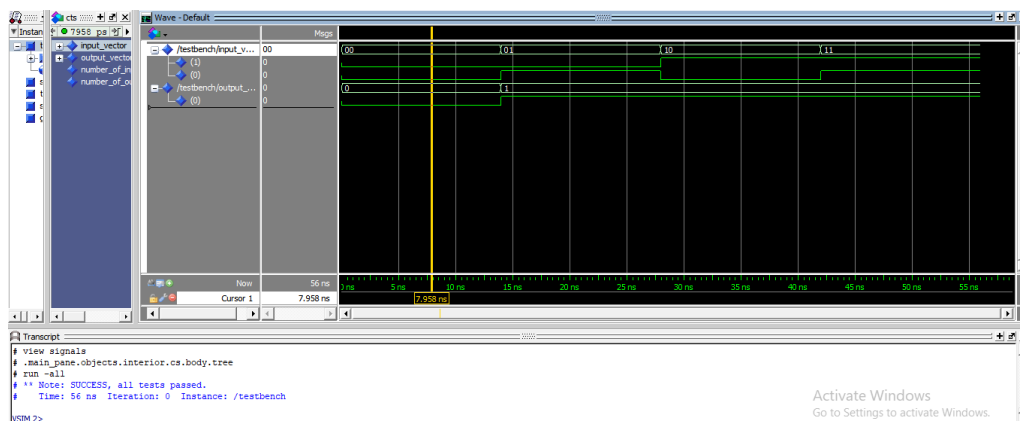


Figure 2: OR gate simulation waveform

XOR gate

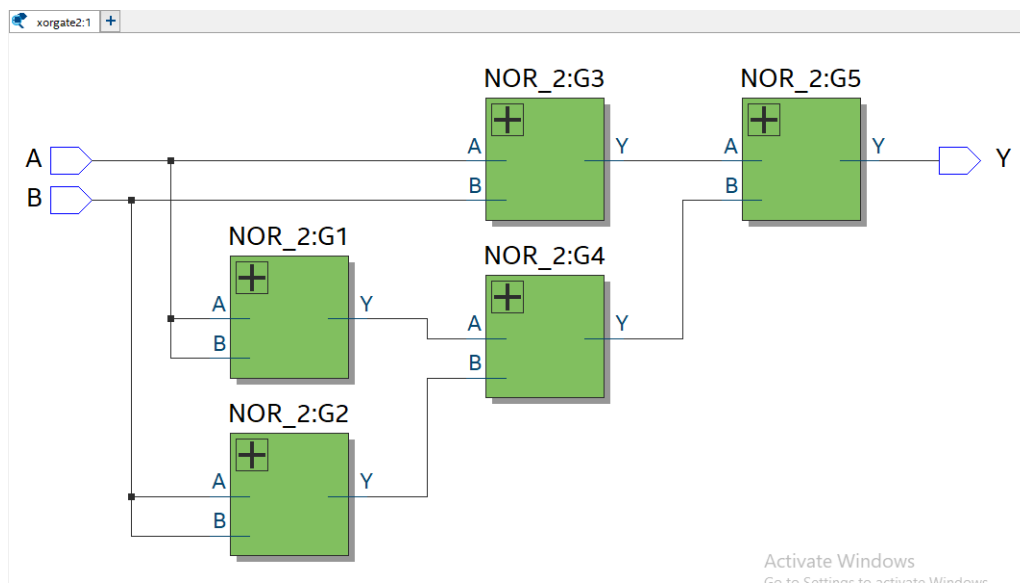


Figure 3: XOR gate schematic

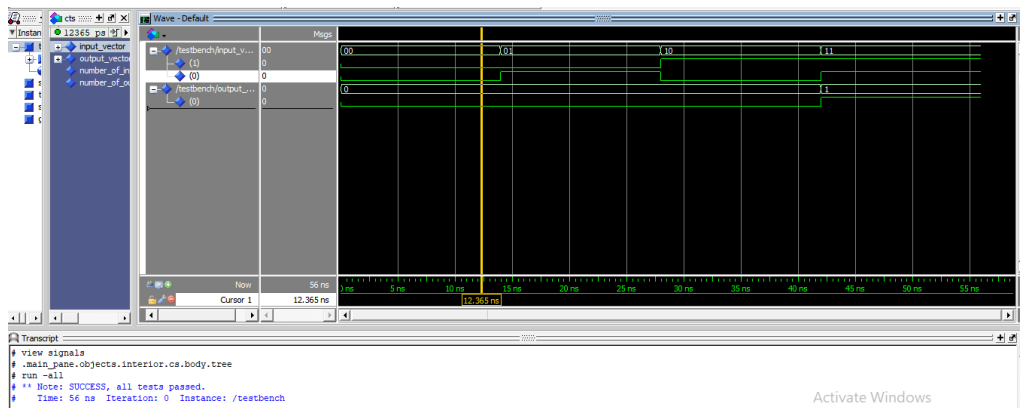


Figure 4: XOR gate simulation waveform

AND gate

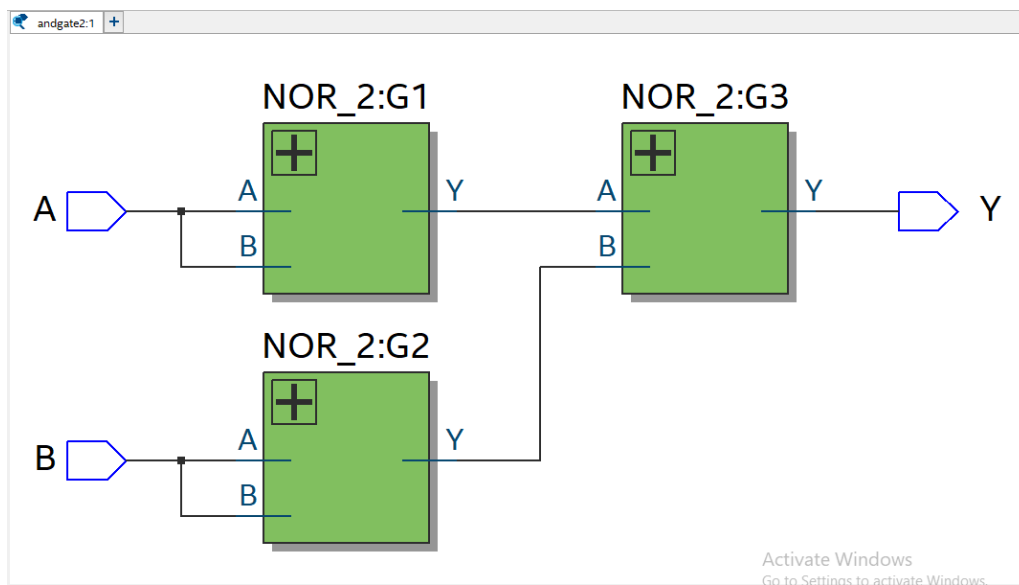


Figure 5: AND gate schematic

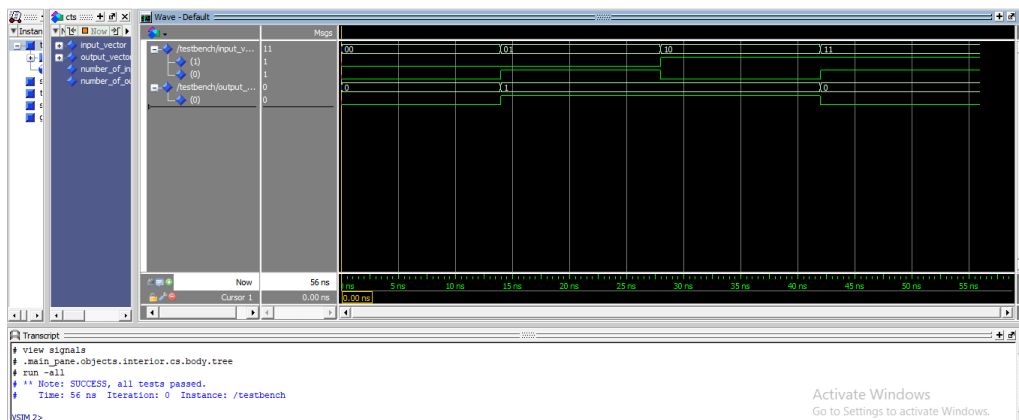


Figure 6: AND gate simulation waveform



Figure 7: Half Subtractor schematic



Figure 8: Half Subtractor simulation waveform

Full Subtractor

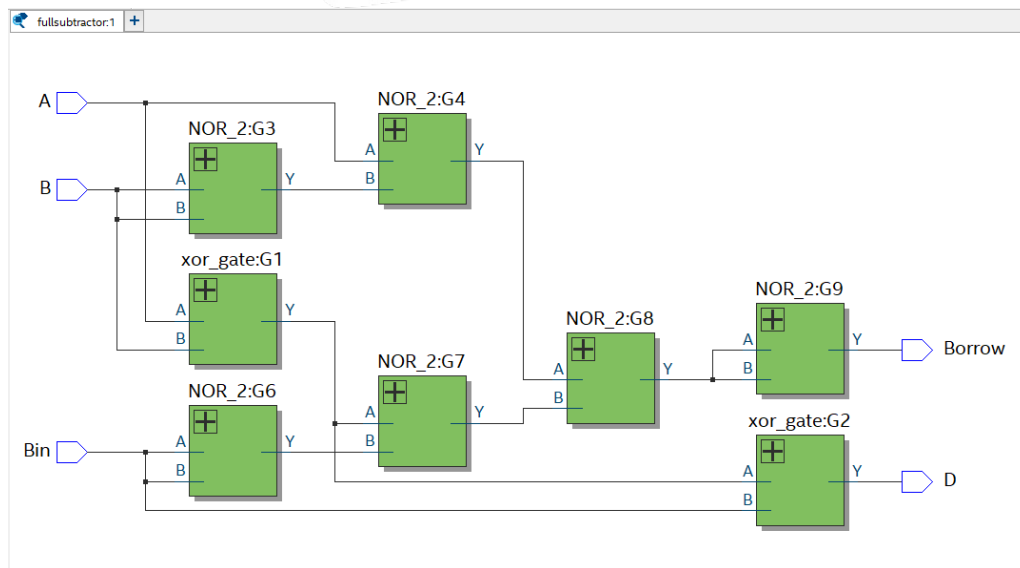


Figure 9: Full Subtractor schematic

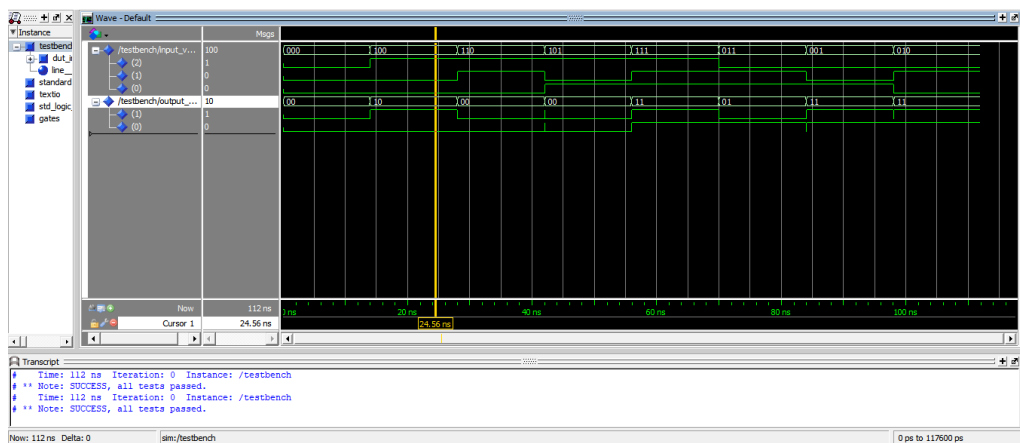


Figure 10: Full Subtractor simulation waveform